

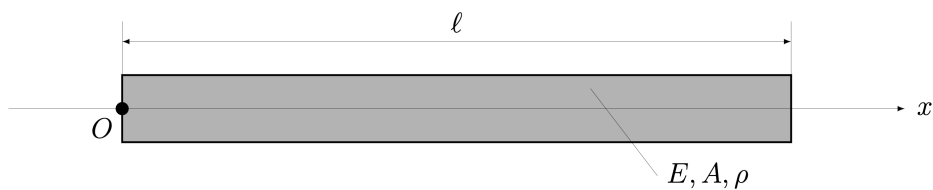


ME-473: DYNAMIC FINITE ELEMENT ANALYSIS OF STRUCTURES

Mini-project 1 report - Natural frequencies and mode shapes of a uniform bar

GROUP F

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1 Introduction

This document presents the analytical and numerical solutions obtained in the study of the deformation and vibration modes of a beam clamped at both ends. The objective is to analyze the beam's vibrational behavior and deformation characteristics using the Finite Element Method (FEM). In our case (Group F), the analysis was performed on cubic elements with clamped boundary conditions at $x = 0$ and $x = l$.

To carry out the numerical analysis, MATLAB was used as the primary software. The study aims to compare the numerical results with analytical solutions to assess the accuracy and effectiveness of the FEM approach in modeling beam vibrations.

2 Methodology

In this case, a modal analysis will be conducted to examine the longitudinal vibrations of the beam. The beam in question is defined by the following parameters:

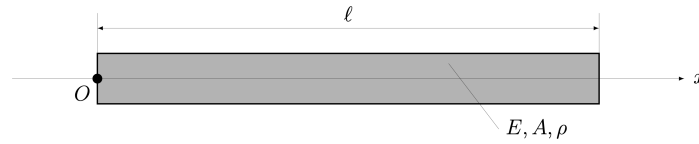


Figure 1: Schematic of the problem setup

- Length: $l = 1$ m
- Cross-sectional area: $A = 0.01$ m²
- Elastic modulus: $E = 210$ GPa
- Density: $\rho = 7850$ kg/m³

The cross-sectional area, elastic modulus, and density are assumed constant. The problem is assumed to be in one dimension, so u describes the deformation in x . The bar is clamped at both ends, then the boundary conditions are $u(x = 0, t) = u(x = l, t) = 0 \forall t \in]0, T[$. Thus, the strong form formulation for longitudinal vibration in one dimension is:

$$\begin{cases} EA \partial_{xx} u = \rho A \ddot{u} & , x \in]0, l[\text{ and } t \in]0, T[\\ u(0, t) = 0 & , t \in]0, T[\\ u(l, t) = 0 & , t \in]0, T[\end{cases}$$

2.1 Analytical Solutions

Given the strong form of the longitudinal vibration of a beam defined above:

$$EA \partial_{xx} u = \rho A \ddot{u} \quad \xrightarrow[\text{A is constant}]{\quad\quad\quad} \boxed{E \partial_{xx} u = \rho \ddot{u}}$$

Suppose that we have a solution $u(x, t) = X(x) \cdot T(t)$, we can implement it within our general equation:

$$EX''(x)T(t) = \rho X(x)T''(t) \implies \frac{\rho}{E} \frac{T''(t)}{T(t)} = \frac{X''(x)}{X(x)} = -\lambda^2$$

where λ is a constant. Therefore, we obtain two different differential equations, with boundary conditions $u(0, t) = u(l, t) = 0$, so $X(0) = X(l) = 0$:

$$\begin{cases} X''(x) + \lambda^2 X(x) = 0 & , x \in]0, l[\\ T''(t) + \lambda^2 \frac{E}{\rho} T(t) = 0 & , t \in]0, T[\\ X(0) = X(l) = 0 \end{cases}$$

From this we obtain the general solution $X(x) = A \sin(\lambda x) + B \cos(\lambda x)$. Knowing the BC's for a clamped-clamped beam, we have that $B = 0$ and $\lambda_n = n\pi/l$. Thus, the natural frequencies of the system can be deduced as:

$$\boxed{\omega_n = \sqrt{\lambda^2 \frac{E}{\rho}} = n\pi \sqrt{\frac{E}{\rho l^2}} \quad n = 1, 2, 3, \dots}$$

Looking at the first four natural frequencies:

$$\begin{cases} \omega_1 = \pi \sqrt{\frac{E}{\rho l^2}} \approx 16'249 \text{ rad/s} \\ \omega_2 = 2\pi \sqrt{\frac{E}{\rho l^2}} \approx 32'498 \text{ rad/s} \\ \omega_3 = 3\pi \sqrt{\frac{E}{\rho l^2}} \approx 48'747 \text{ rad/s} \\ \omega_4 = 4\pi \sqrt{\frac{E}{\rho l^2}} \approx 64'996 \text{ rad/s} \end{cases} \quad (2.1.1)$$

2.2 Numerical Simulations

Given the strong form shown in the section above we are able to get to the weak form through:

$$\begin{aligned} \int_0^L EA \partial_{xx} u \delta u \, dx &= \int_0^L \rho A \ddot{u} \delta u \, dx \\ [EA \partial_x u \delta u]_0^L - \int_0^L EA \partial_x u \partial_x \delta u \, dx &= \int_0^L \rho A \ddot{u} \delta u \, dx \end{aligned} \quad (2.2.1)$$

With $u(0, t) = 0$ and $u(L, t) = 0$ we get $\delta u(0, t) = 0$ and $\delta u(L, t) = 0$. Which leads to the weak form:

$$\int_0^L EA \partial_x u \partial_x \delta u \, dx + \int_0^L \rho A \ddot{u} \delta u \, dx = 0 \quad (2.2.2)$$

To get the local form of the equation, we get an approximation of the u : ${}^e u^h = {}^e H q$ and ${}^e \delta u^h = {}^e H \delta q$. From this we get the local mass and stiffness matrices:

$$\begin{aligned} {}^e K_{ij} &= \int_0^L {}^e E {}^e A \frac{\partial {}^e h_i}{\partial x} \frac{\partial {}^e h_j}{\partial x} dx \\ {}^e M_{ij} &= \int_0^L {}^e \rho {}^e A {}^e h_i {}^e h_j dx \\ \partial q^T [M \ddot{q}(t) + K q(t)] &= 0 \end{aligned} \quad (2.2.3)$$

Finally, a change of variable is performed with $\xi = 1/l = (2x - x_i - x_{i+2})$ which gives $\frac{d\xi}{dx} = \frac{2}{l}$. This gives a local stiffness and mass matrix of the form:

$$\begin{aligned} {}^e K_{ij} &= \frac{2 {}^e E {}^e A}{l} \int_{-1}^1 \frac{\partial {}^e h_i}{\partial \xi} \frac{\partial {}^e h_j}{\partial \xi} d\xi \\ {}^e M_{ij} &= \frac{l {}^e \rho {}^e A}{2} \int_{-1}^1 {}^e h_i {}^e h_j d\xi \end{aligned} \quad (2.2.4)$$

This is the equation that is then computed and assembled, with a different number of element by the MATLAB algorithm to find the different modes.

3 Results and Discussion

This section presents the results obtained from our analysis. First, we will present the results obtained using the analytical approach, and then the numerical results obtained with MATLAB.

3.1 Results and comparison

Mode	Analytical Solution	Numerical Results				
		$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$
ω_1	16'249	16'356	16'250	16'249	16249	16'249
ω_2	32'498	33'520	32'712	32'500	32'498	32'498
ω_3	48'747	N/A	50'030	48'781	48'747	48'747
ω_4	64'996	N/A	67'039	65'424	65'000	64'996

Table 1: Summary table of the natural frequencies obtained, all values are given in **rad/s**.

The Table 1 above clearly shows a convergence from the numerical solution towards the analytical solution, to see this more clearly, the error between both solutions has been calculated.

The error is computed as:

$$\delta_{i,n} = \frac{\omega_{i,n} - \omega_{i,exact}}{\omega_{i,exact}}$$

Mode	Error of Numerical Results [-]				
	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$
ω_1	6.584e-03	6.830e-05	1.139e-06	1.809e-08	2.830e-10
ω_2	3.144e-02	6.584e-03	6.830e-05	1.139e-06	1.809e-08
ω_3	N/A	2.633e-02	6.950e-04	1.263e-05	2.047e-07
ω_4	N/A	3.144e-02	6.584e-03	6.830e-05	1.139e-06

Table 2: Summary table of the natural frequencies errors computed, all values are dimensionless.

The Table 2 here shows how the error between the analytical and numerical solution, showing it decreasing as the number of element increases. To show this more clearly, the error and the norm of the error are plotted respectively in Figure 2 and 3 below. The norm of the error vector can also be defined as:

$$\delta_n = \sqrt{(\delta_{1,n})^2 + \dots + (\delta_{m,n})^2}$$

with n the number of elements and m the chosen number of modes, which is 2 for $n = 1$, and 4 for $n \geq 2$.

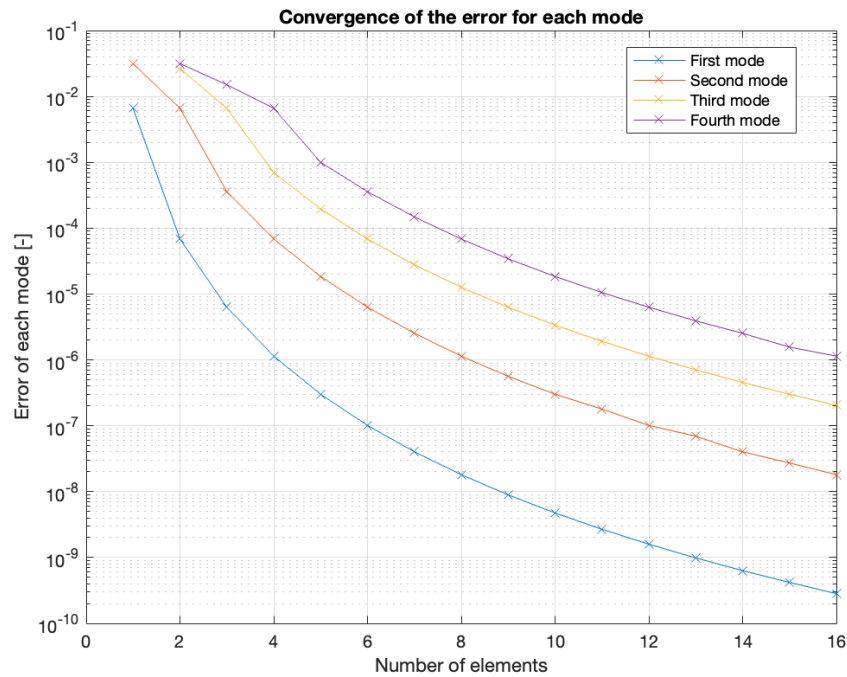
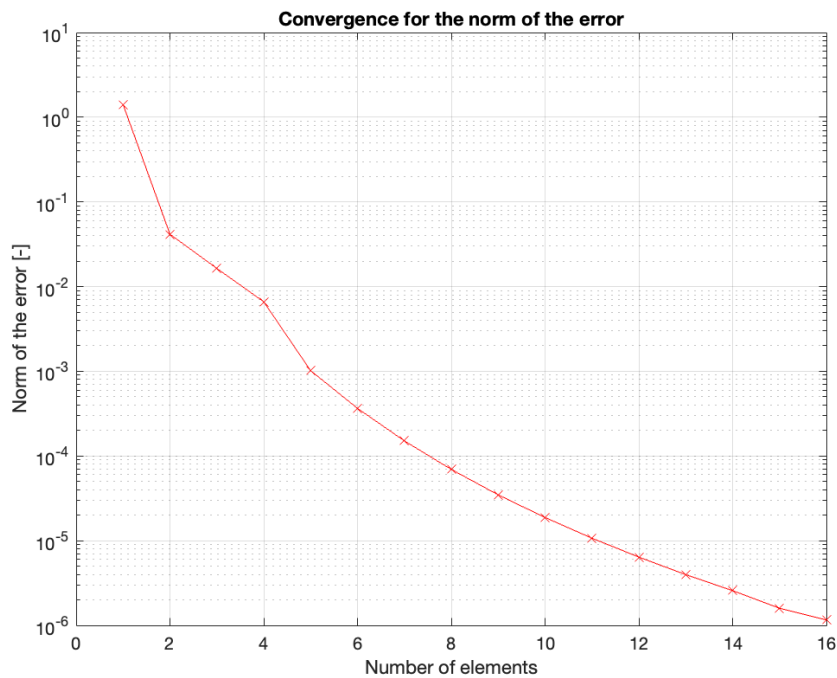


Figure 2: Convergence of each mode with increasing number of elements

Figure 3: Convergence of the norm of the error δ_n

3.2 Mode Shapes

The mode shapes of a clamped-clamped beam describe its deformation patterns at specific natural frequencies. These shapes are obtained from the analytical solution of the beam's governing equation, considering the boundary conditions. In Figure 4 below, the first four analytical mode shapes are plotted to illustrate the characteristic vibration patterns of the beam. Furthermore, these analytical results are validated from other paper sources [3].

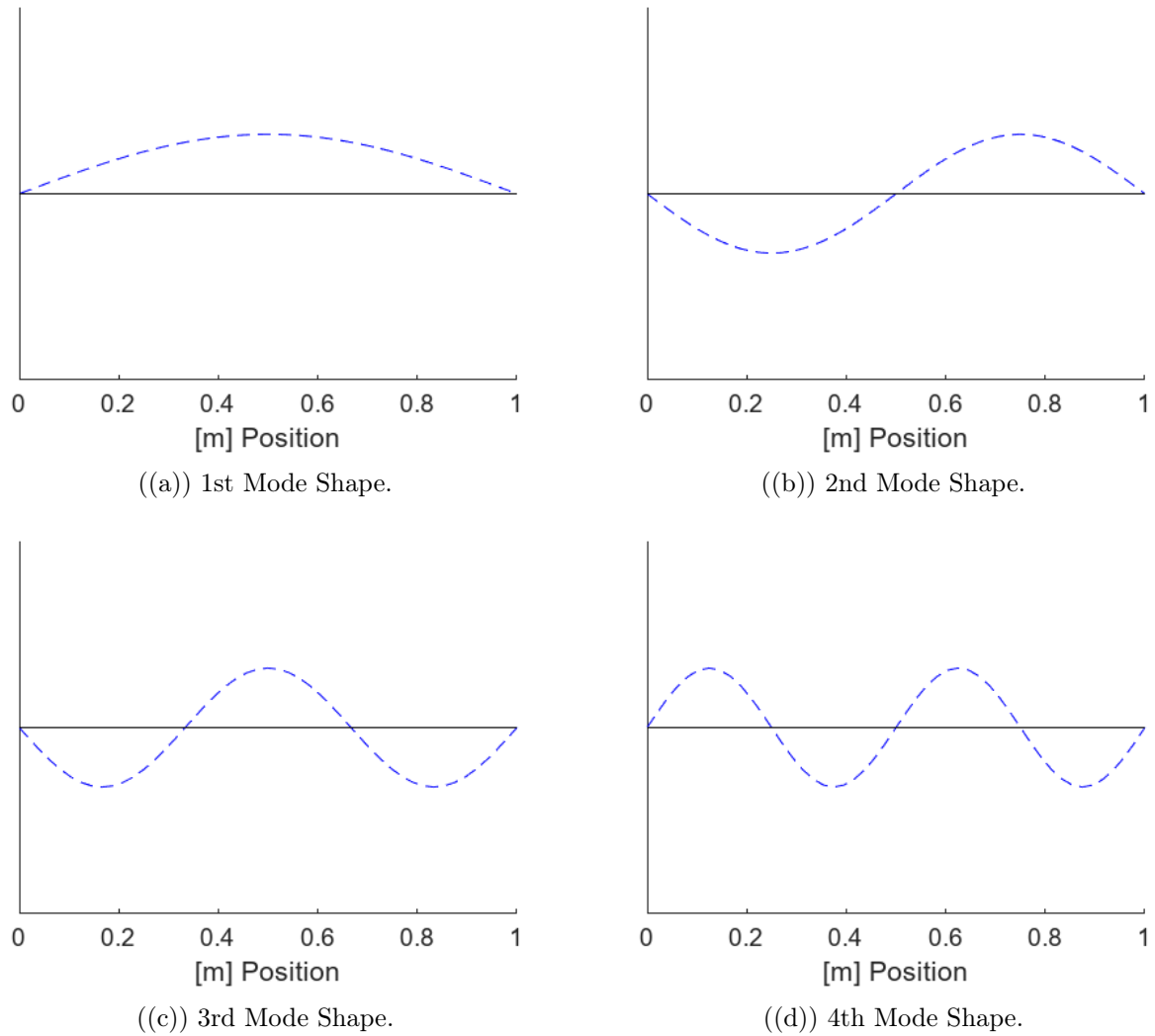


Figure 4: First four mode shapes of the clamped-clamped beam.

Figure 5 shows the second mode shape of a clamped-clamped beam for different mesh refinements. With fewer elements, the numerical solution deviates from the analytical mode shape, but as the number of elements increases, the approximation improves. This demonstrates the expected convergence of the numerical model. Additional plots are provided in the appendices A, B, C and D.

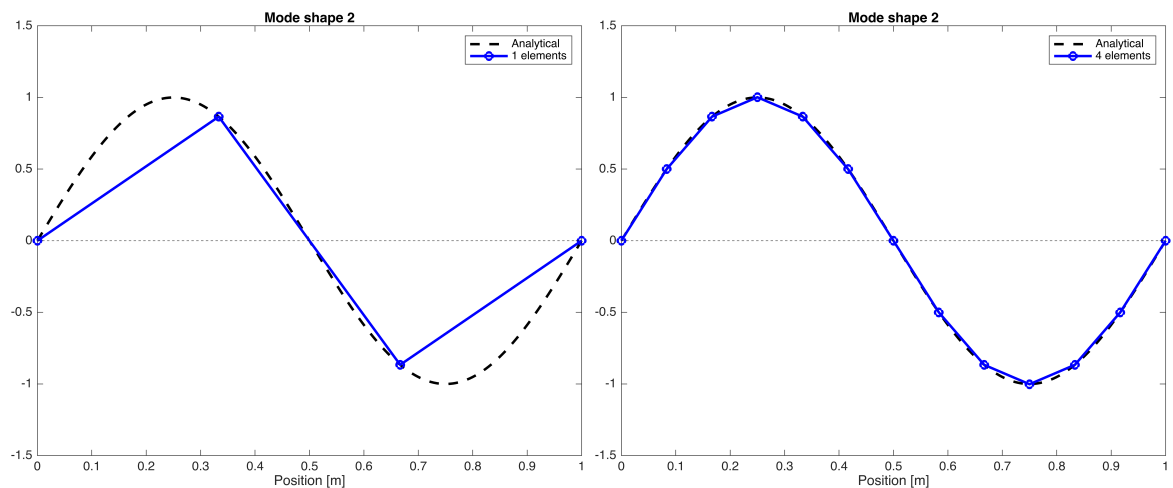


Figure 5: Comparison between the analytical, in dashed black, and the numerical, in blue, second mode shape for increasing number of elements. The mode shapes are amplified.

3.3 Discussion

The results demonstrate the expected convergence of the numerical solution toward the analytical mode shapes and natural frequencies as the number of elements increases. The error analysis in Table 2 and Figure 2 confirms that the accuracy improves significantly with finer mesh refinement. Looking at Figure 6 below we see that, by doubling the number of elements, the error is divided by approximately 2 decades, or around 100. This shows the effectiveness of using cubic shape function to get a correct answer.

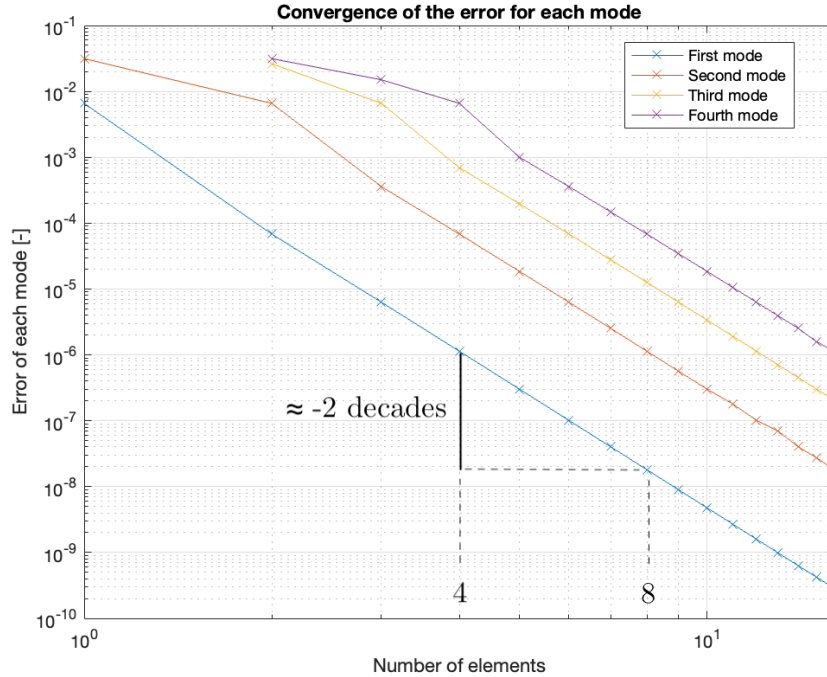


Figure 6: Logarithmic plot of the error of each mode

The mode shape comparison (Figure 5) highlights how a coarse mesh leads to significant deviations from the analytical solution, while a finer mesh closely follows the expected deformation pattern. This emphasizes the importance of choosing an adequate number of elements to achieve reliable results without excessive computational cost.

Overall, these findings validate the numerical approach and confirm that increasing mesh refinement leads to improved accuracy, following the theoretical expectations of finite element methods.

4 Conclusion

In this study, we successfully analyzed the natural frequencies and mode shapes of a clamped-clamped beam using both analytical methods and numerical simulations through MATLAB. The comparison between the analytical and numerical results demonstrated a strong correlation, confirming the validity and accuracy of the Finite Element Method (FEM) for modeling the vibrational behavior of the beam. Minor discrepancies observed in higher modes can be attributed to the discretization limitations inherent to numerical modeling. Future improvements could include refining the mesh further or exploring more advanced numerical techniques to minimize errors. Additionally, extending the analysis to include damping effects or complex boundary conditions could provide deeper insights into the beam's vibrational characteristics.

References

- [1] Stefano F. Burzio. (Spring 2025). Lecture Notes: *ME-473: Dynamic finite element analysis of structures*. EPFL.
- [2] Thomas Gmür. (2014). *Dynamique des structures*. Presses polytechniques et universitaires romandes.
- [3] Yüksel, Şefaaddin & Dalli, Uğur. (2005). *Longitudinally Vibrating Elastic Rods with Locally and Non-Locally Reacting Viscous Dampers*. Shock and Vibration. 12. 109-118. 10.1155/2005/747089.

A Mode 1

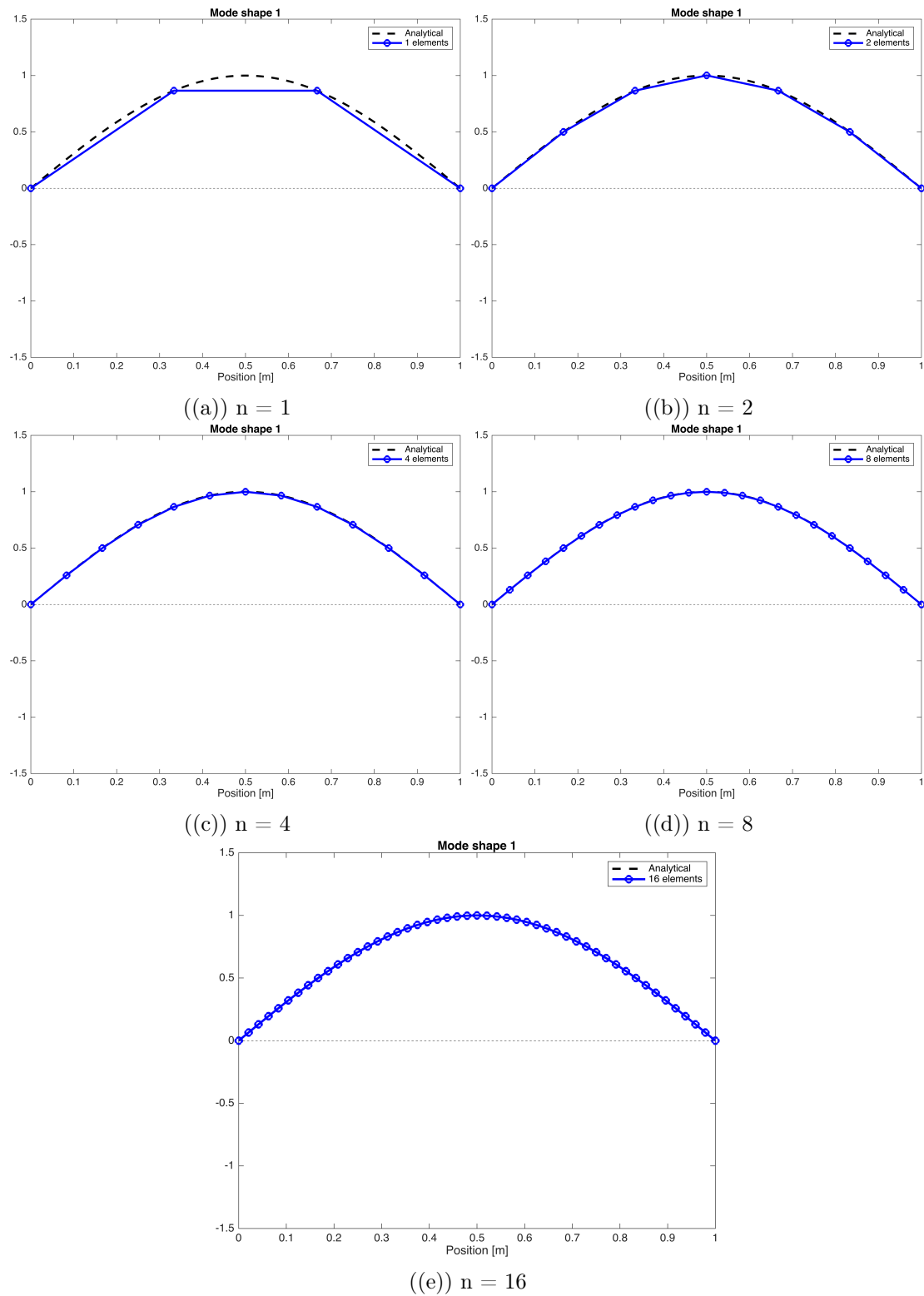


Figure 7: FEM Analysis mode 1

B Mode 2

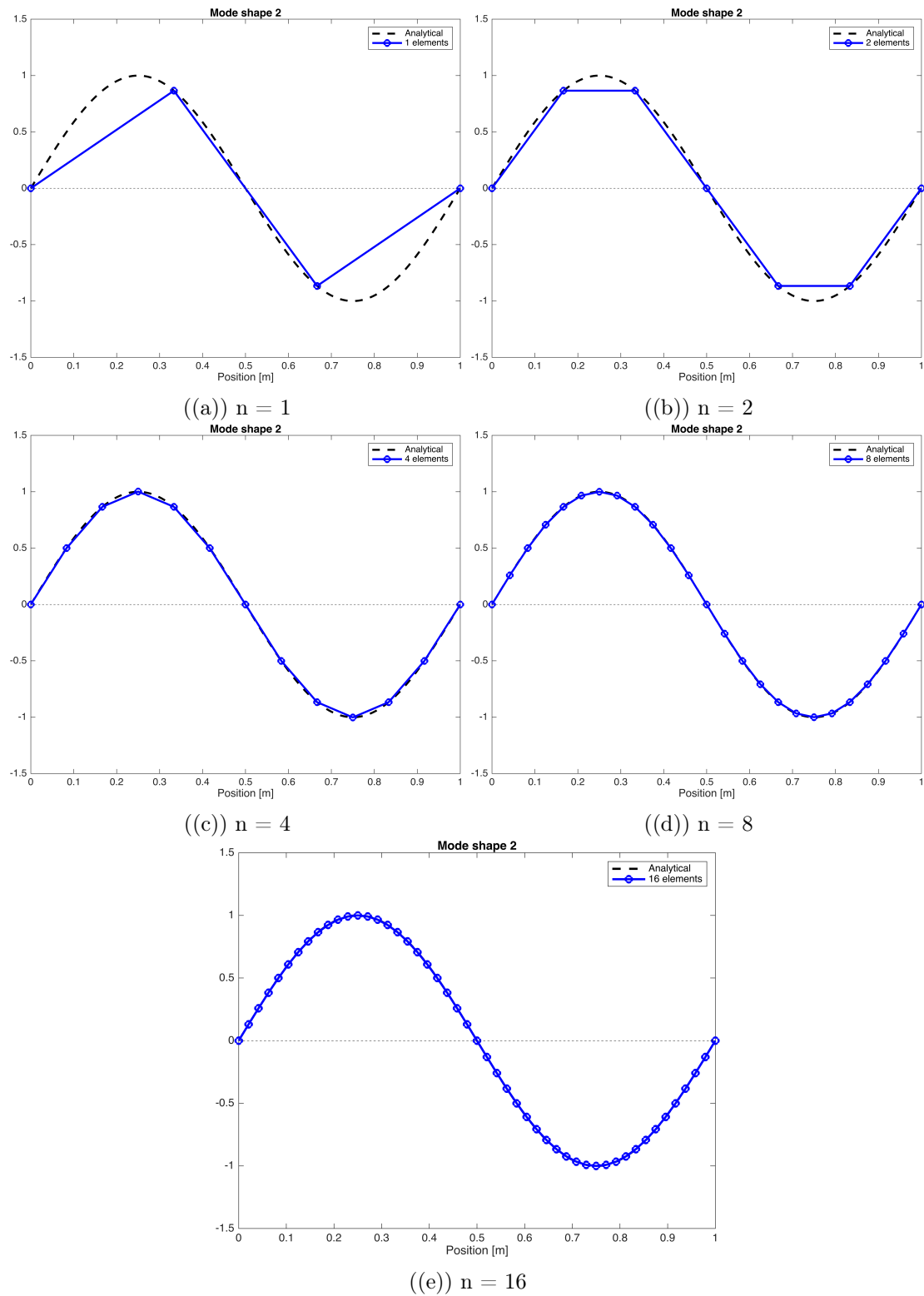


Figure 8: FEM Analysis mode 2

C Mode 3

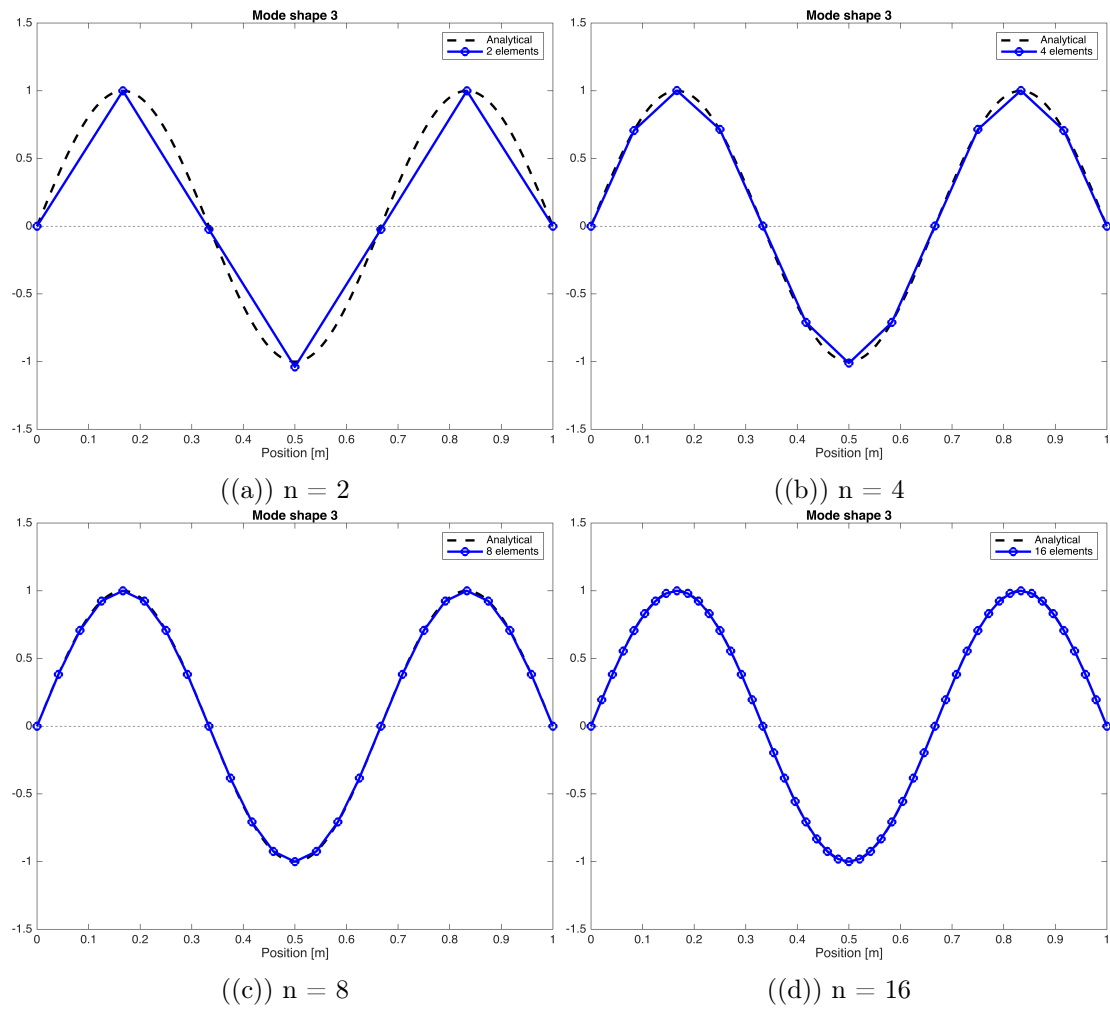


Figure 9: FEM Analysis mode 3

D Mode 4

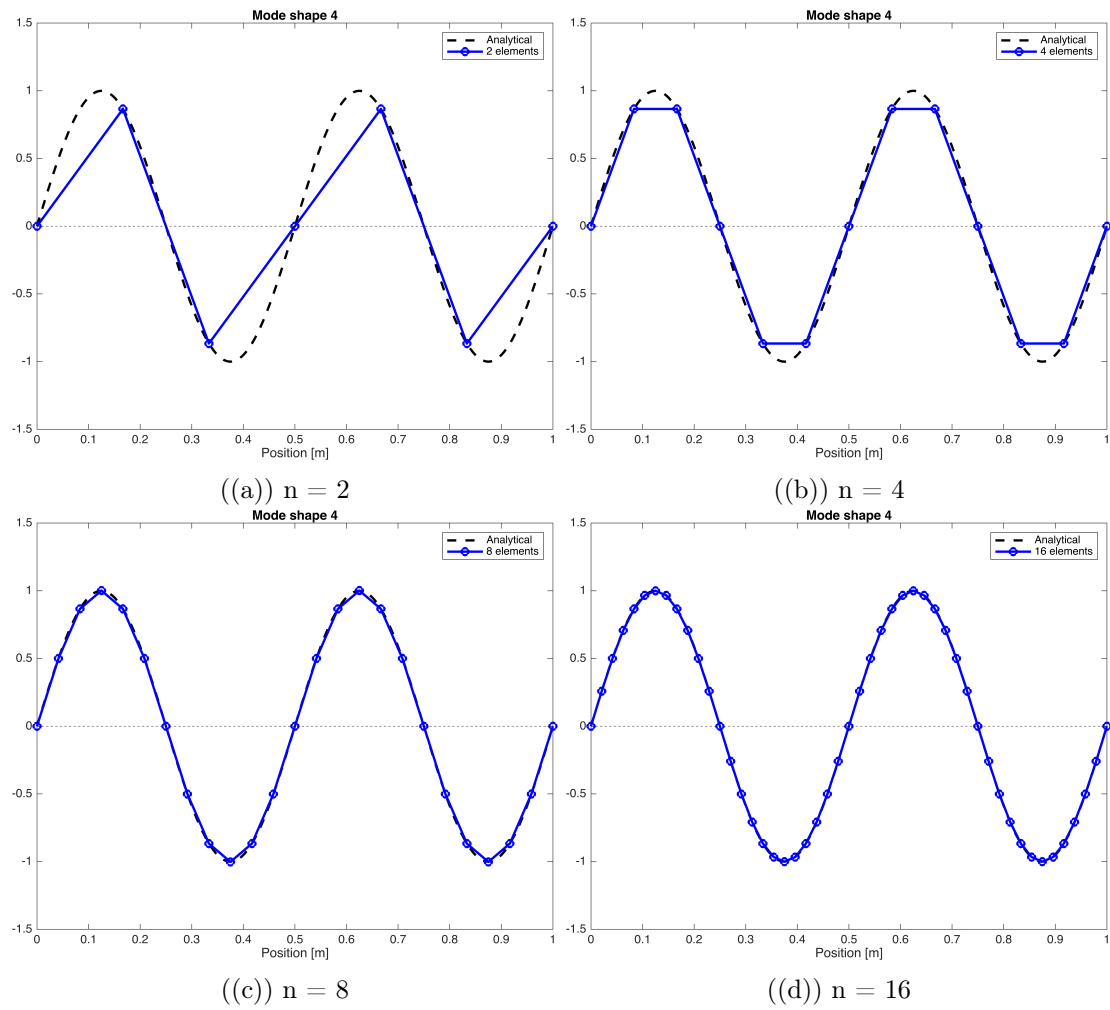


Figure 10: FEM Analysis mode 4